# 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-TELEMETRY-C-EN · v1.5 · 2026-10-20 |
| Package | star-aos-c |
| Version | 0.1.3 |
| Capability | star.telemetry.aos-c |
| Maturity | VERIFIED |
| Python | ⩾3.10 |
| Source commit | 94fcf7c |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_telemetry_c |

# 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

# 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

# 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_aos_c-0.1.3-py3-none-any.whl
```

| FILE | SHA-256 |
|---|---|
| star_aos_c-0.1.3-py3-none-any.whl | 192d0b6eac36ed5cb6984bd606957901d3be024b3c94a60823ac01acb3ff1089 |
| star_telemetry_c-0.1.3-source.zip | a9567b095b7fad0986743d4dd2d77a4ea73406ec112d5ed1c540da500f5bb82a |

# 5. What it does, and its limits

CCSDS AOS v2 / TM v1 primary header decode and FECF CRC-16 check in C99, verified identical on 18 emulated ISAs and 12 simulated MCUs / space processors, cross-checked with spacepackets, fuzzed and mutation-tested.

No timing (DWT is not modelled), no radiation behaviour, no flight heritage; simulated and emulated targets only. Defects found by this verification and fixed: TM VC count read from the master-channel octet, OCF read as "replay", OCF and secondary header left in the data (2026-10-04).

- `star_aos_parse(frame, len, has_fecf, &hdr)`: TFVN, SCID, VCID, VC frame count, replay (AOS), FHP; TM: VC count from octet 3, OCF flag (4 octets before the FECF), secondary header skipped (length from its ID octet).
- FECF: CRC-16/CCITT-FALSE over the frame without the FECF; `fecf_valid` only when computed == received.
- Exact layout: `data_offset + data_len + ocf_len (+2 with FECF) == len`; TM data offset 6 or 7 + (ID & 0x3F).
- Errors: NULL inputs → `STAR_AOS_ERR_NULL`; frames too short for the declared layout → `STAR_AOS_ERR_SHORT`; never a read outside `[frame, frame + len)`.
- Portability: identical results on every target listed below (byte-wise parsing, no undefined behaviour).

# 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

# 7. Citation and bill of materials

```
@software{cavazzini_star_telemetry_c_0_1_3,
  author  = {Cavazzini, Andrea},
  title   = {{star-aos-c}: CCSDS AOS/TM transfer-frame check in C99 (S.T.A.R.)},
  version = {0.1.3},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_telemetry_c/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

# 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts